

Supplementary Information

Bacterial persistence promotes the evolution of antibiotic resistance by increasing survival and mutation rates

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Supplementary Methods

Mathematical model of antibiotic treatment of an *in vivo* bacterial infection

Differential equation system

To investigate the implications of the persistence-resistance link on the *in vivo* clearance of bacterial infections, we used a mathematical model described by the following differential equation system:

$$N'_W(t) = g_W(A) N_W(t) - a N_W(t) + b P_W(t) - \mu_W N_W(t) - j_N N_W(t) E(t)$$

$$P'_W(t) = a N_W(t) - b P_W(t) - \mu_P P_W(t) - j_P P_W(t) E(t)$$

$$N'_{R1}(t) = g_{R1}(A) N_{R1}(t) - a N_{R1}(t) + b P_{R1}(t) + \mu_W N_W(t) - \mu_{R1} N_{R1}(t) - j_N N_{R1}(t) E(t)$$

$$P'_{R1}(t) = a N_{R1}(t) - b P_{R1}(t) + \mu_P P_W(t) - \mu_P P_{R1}(t) - j_P P_{R1}(t) E(t)$$

$$N'_{R2}(t) = g_{R2}(A) N_{R2}(t) - a N_{R2}(t) + b P_{R2}(t) + \mu_{R1} N_{R1}(t) - \mu_{R2} N_{R2}(t) - j_N N_{R2}(t) E(t)$$

$$P'_{R2}(t) = a N_{R2}(t) - b P_{R2}(t) + \mu_P P_{R1}(t) - \mu_P P_{R2}(t) - j_P P_{R2}(t) E(t)$$

$$N'_{R3}(t) = g_{R3}(A) N_{R3}(t) - a N_{R3}(t) + b P_{R3}(t) + \mu_{R2} N_{R2}(t) - j_N N_{R3}(t) E(t)$$

$$P'_{R3}(t) = a N_{R3}(t) - b P_{R3}(t) + \mu_P P_{R2}(t) - j_P P_{R3}(t) E(t)$$

$$E'(t) = q (e_{\max} - E(t)) - l E(t)$$

where N_W and P_W represent the size of the wild type populations of normal and persister cells, N_{R1} , N_{R2} , and N_{R3} are the size of the population of first-, second-, and third-step resistant mutants (resistant to exposure to increasing antibiotic concentrations), P_{R1} , P_{R2} , and P_{R3} are the corresponding number of persister cells of the resistant populations, and E represents the number of activated effector cells of the innate immune response at the infection site. The other symbols represent the growth or killing rate (g), persister formation (a) and resuscitation (b) rates, mutation rates (μ), and immune-mediated killing rates (j). The different elements of the model are further explained below.

Persister formation

As in other models, the transition between normal and persister cell states is governed by the forward and backward switching rates a and b ¹⁻⁴. To simulate increased switching to the persister state during late-exponential and stationary phase, a is made proportional to the cell density via $a(t) = a_{\max} \times \frac{N_{\text{tot}}(t)}{c}$, with $a_{\max}$ the maximum switching rate during stationary phase, $N_{\text{tot}}(t)$ the total bacterial population size at time t , and c the carrying capacity³. Switching rates are assumed to be equal for the wild type and resistant populations. For the baseline situation, we chose $a_{\max} = 0.0001$ and $b = 0.1$, in line with values that have repeatedly been found experimentally for wild type *E. coli* strains^{1,5}.

Growth and killing

Growth or killing of all non-persister populations (N_w , N_{R1} , N_{R2} , and N_{R3}) is described with a pharmacodynamic Hill function, which relates the net bacterial growth or death rate g to the antibiotic concentration A ⁶:

$$g(A) = g_{\max} - \frac{(g_{\max} - g_{\min}) \times \left(\frac{A}{\text{MIC}}\right)^\kappa}{\left(\frac{A}{\text{MIC}}\right)^\kappa + \frac{g_{\min}}{g_{\max}}}$$

where $g_{\min}$ and $g_{\max}$ represent the minimum and maximum growth rate, and MIC the minimum inhibitory concentration. The Hill coefficient κ describes the sensitivity of the bacterial growth rate to changes in antibiotic concentration. We used $g_{\max} = 1 \text{ h}^{-1}$, $g_{\min} = -6.5 \text{ h}^{-1}$ and $\kappa=1$ according to experimentally determined values for *E. coli* ⁶.

To incorporate logistic growth, $g(A)$ is multiplied by $\left(1 - \frac{N_{\text{tot}}}{c}\right)$ during growth. For persister cells, $g(A)$ equals zero, as we assume that they do not grow and are not killed by antibiotics. Since MIC values differ for populations with different resistance levels, we specifically define $g(A)$ for wild type cells and the three types of resistant mutants as $g_w(A)$, $g_{R1}(A)$, $g_{R2}(A)$, and $g_{R3}(A)$ respectively.

Antibiotic treatment

The antibiotic $A(t)$ is administered periodically every t_{peak} hours at a peak concentration of $A_{\max}$, after which the concentration at the infection site declines exponentially at rate w : $A'(t) = -w \times A(t)$.

This kind of antibiotic profile mimics periodic intravenous bolus injections, or oral administration when absorption kinetics of the drug are fast and can be ignored. We set $A_{\max} = 25 \text{ } \mu\text{g/ml}$ and $t_{\text{peak}} = 12 \text{ h}$. We further choose $w=0.277259$, so that the half-life of the antibiotic, $t_{1/2} = \frac{\ln(2)}{w}$ equals 2.5 h. For the wild type population with MIC=1 $\mu\text{g/ml}$, this gives the following pharmacodynamic parameters:

- peak/MIC-ratio of 25
- 24-h AUC/MIC-ratio of 173.9 ($2 \times \int_0^{12} A_{\max} e^{-wt} = \int_0^{12} 25 e^{-0.277259t} = 173.9$)
- $t > \text{MIC}$ of 96.75% per dosing interval ($25 e^{-0.277259t} = 1 \Leftrightarrow t = 11.6096$; $\frac{11.6096}{12} = 0.9675$)

These pharmacokinetic and pharmacodynamic parameters are within a clinically realistic and favorable range for treatment with many antibiotics including the fluoroquinolones and aminoglycosides ⁷.

Mutation rates and the evolution of genetic resistance

Because single mutations conferring clinical resistance are rare, resistance evolution is modelled as a step-wise process, with subsequent mutations generating mutants with gradually increasing MIC values (N_{R1} , N_{R2} , and N_{R3}) ⁸. Resistance-conferring mutations in non-persister cells arise at a rate $\mu(t)$:

$$\mu(t) = m g(A) + \tau m e^{-h g(A)}.$$

In the first term, m is the mutation rate per division in optimal growth conditions, while the second term describes the increase in mutation rates observed in the presence of sublethal concentrations of antibiotics. The latter increases with steepness h as the growth rate decreases due to the presence of antibiotics⁹. During optimal growth, when $g(A) \cong g_{\max}$, the second term approximates zero, resulting in $\mu = m g(A)$. When $g(A) = 0$, m equals τm . Thus, τ gives the maximum fold increase of the mutation rate when cells grow in the presence of antibiotics. Mutation rates between all genotypes are taken equal. Since growth rates differ for populations with different resistance levels, we specifically define m for wild type cells and the three types of resistant mutants as μ_W , μ_{R1} , μ_{R2} , and μ_{R3} respectively.

For the baseline situation, we set the mutation rate m for a single resistance-conferring mutation at 1.5×10^{-8} mutations per cell per division and the maximum fold increase due to stress (τ) at 2. These values are in line with our own measurements (see Supplementary Table 5) and values reported by others¹⁰. All mutation rates of growing cells are set equal to zero during killing. For strains with increased mutation rates, we double the mutation rate m , roughly in agreement with our results (see Fig. 3 of the main text and Supplementary Table 5).

Resistance-conferring mutations in persister cells arise at a constant, growth-independent rate μ_p . According to the results of Cirz et al. (2005)¹¹, this mutation rate is 10^4 -fold higher than the spontaneous mutation rate, thus we set $\mu_p = 1.5 \times 10^{-4}$. For high-persistence strains with associated increased mutation rates in Fig 4b of the main text, we take a fourfold higher mutation rate, in agreement with the minimal effect observed in our experiments (see strain *oppB** in Fig. 3b of the main text). The resulting mutants (P_{R1} , P_{R2} , and P_{R3}) remain in the persister state until they start growing due to phenotypic switching to N_{R1} , N_{R2} or N_{R3} , respectively.

Innate immune response

According to Ankomah and Levin (2014), we incorporate the action of a pathogen density-independent innate immune response caused by activated effector cells $E(t)$. Effector cells are recruited to the infection site at a rate proportional to a constant q and the density of effector cells in the reservoir, $e_{\max} - E(t)$, with $e_{\max}$ the total density of effector cells in the reservoir. Activated effector cells are inactivated at a constant rate l . The killing rate of bacteria by immune cells is proportional to the density of immune cells and a killing parameter j_N for normal cells or j_P for persister cells. All parameter values were taken from Ankomah and Levin (2014).

Selective advantage of mutants with pleiotropically linked persistence and mutation rates

Invasion fitness, *i.e.* the long-term relative fitness of a mutant relative to the wild-type, under conditions described above, was calculated as:

$$\frac{P_M(t_2) \times (1 - P_M(t_1))}{P_M(t_1) \times (1 - P_M(t_2))}$$

Where $P_M(t)$ represents the frequency of the mutant strain at time t , and t_1 and t_2 represent the time before and after one antibiotic treatment episode. The mutant is taken to be initially rare in the population (*i.e.* one cell).

Supplementary Figures

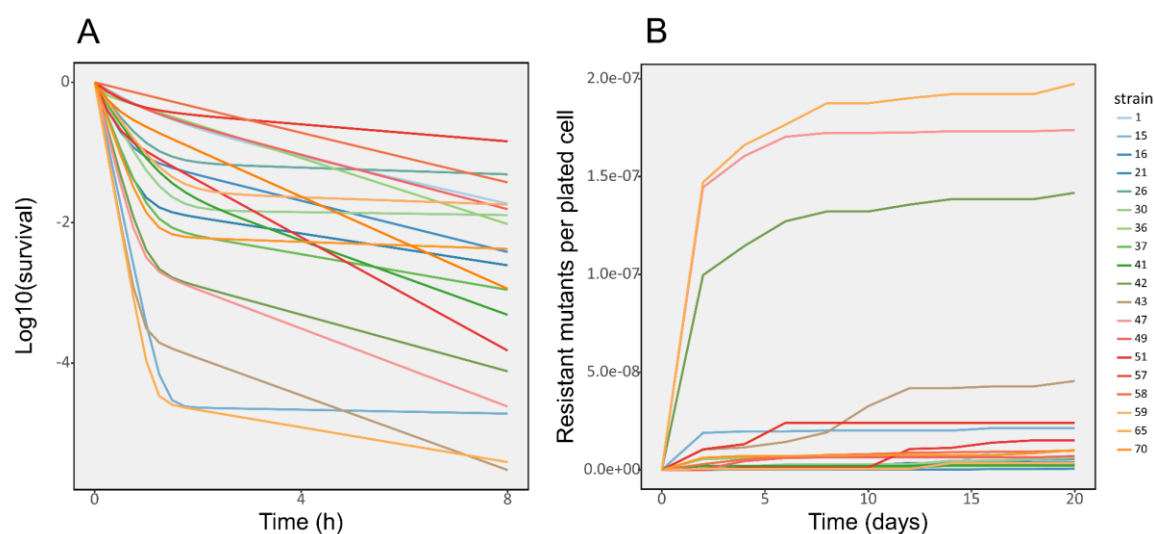


Figure S1. Diversity in persister fractions and resistance development of 20 ECOR isolates. (A) Fittings of biphasic exponential curves to ciprofloxacin time-kill data of exponentially growing ECOR isolates. The average fit on three replicate experiments is shown. Best-fit estimated parameter values are given in Table S2. (B) Cumulative number of resistant colonies per plated cell on ciprofloxacin-containing Mueller-Hinton agar plates. Colonies were counted every two days and averages from 5 replicate plates are shown. Error bars were omitted for clarity. Results for ECOR 8 are not shown because of the large deviation from the rest of the data.

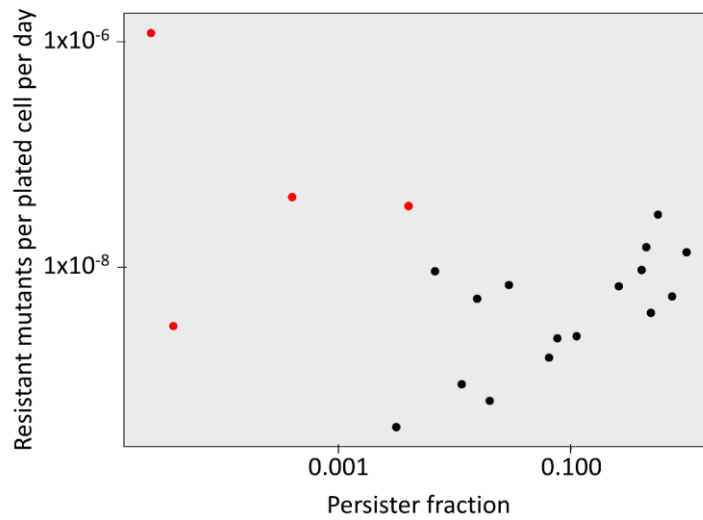


Figure S2. Correlation plot of persister fractions and resistance development of all ECOR strains. Persister fractions (n=3 per strain) and the average number of resistant mutants per plated cell per day (n=5 per strain) on ciprofloxacin-containing Mueller-Hinton agar (4x MIC) are plotted for all 20 ECOR isolates on a log-log scale. These data are equivalent to the data shown in Figure 1 of the main text, except for the four outliers (red points) that were omitted from Figure 1 based on their Cook's distance.

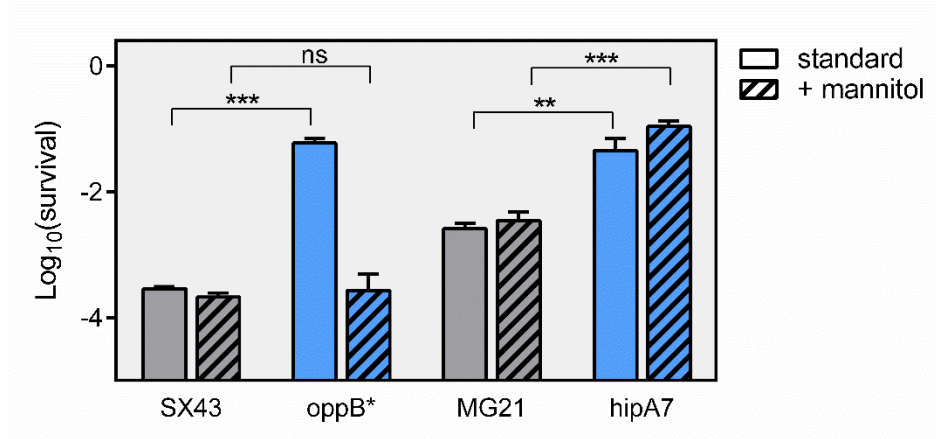


Figure S3. Influence of mannitol on persister fractions of the *oppB and *hipA7* high-persistence mutants.** Survival is plotted for stationary phase cultures treated with a high dose of ciprofloxacin (5 µg/ml) for 5 hours, with or without 1 hour pretreatment with mannitol (10 mM). Error bars represent s.e.m. (n=3). Log₁₀-transformed surviving fractions were compared statistically using unpaired two-sided *t*-tests, after checking equality of variances with F-tests. ***p*≤0.01, ****p*≤0.001, and ns, non-significant.

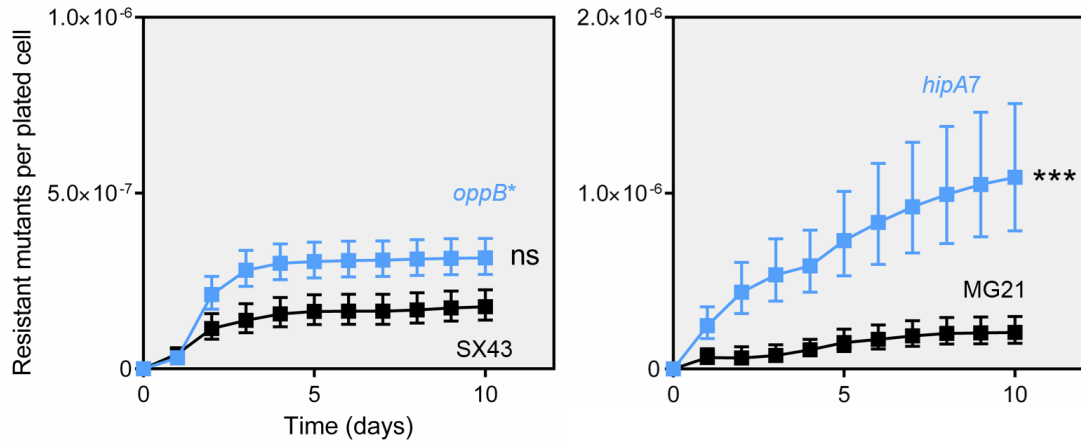


Figure S4. Emergence of antibiotic-resistant colonies in the presence of mannitol. The cumulative number of resistant mutants per plated cell on ciprofloxacin-supplemented Mueller-Hinton agar (2x MIC) containing mannitol (10 mM). Colonies were counted each day and averages from 10 replicate plates are shown. Error bars represent s.e.m. Statistical differences were determined using two-sided *t*-tests on coefficients of a log-linear model. *** $p \leq 0.001$ and ns, non-significant.

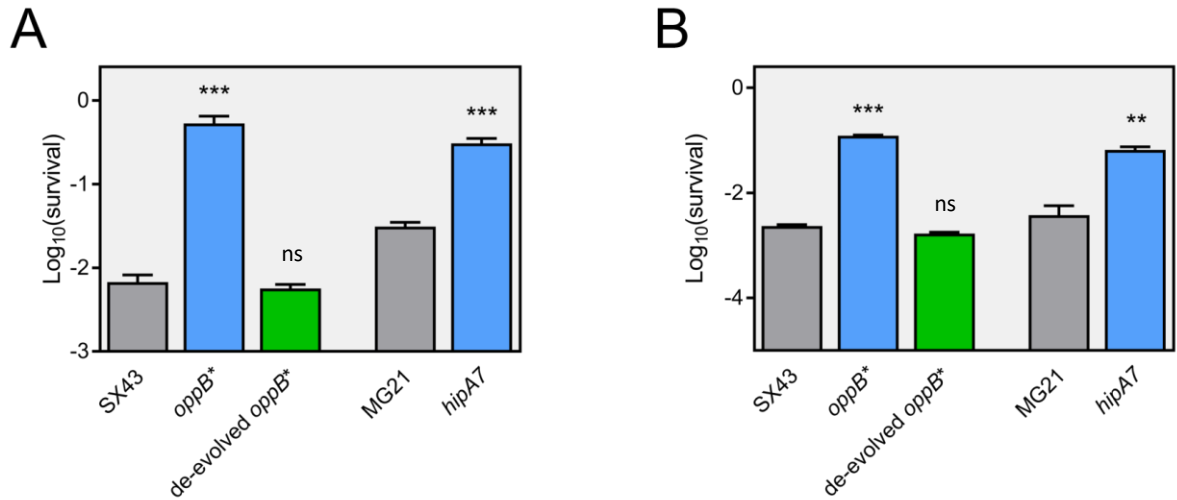


Figure S5. Persister fractions after treatment with norfloxacin and kanamycin. Survival is plotted for stationary phase cultures treated with a high dose of norfloxacin (5 $\mu\text{g/ml}$) (A) or kanamycin (400 $\mu\text{g/ml}$) (B) for 5 hours. Error bars represent s.e.m. (n=3). Log_{10} -transformed surviving fractions were compared statistically using two-sided *t*-tests, with Welch's correction in the case of unequal variances (checked with F-tests). ** $p \leq 0.01$, *** $p \leq 0.001$, and ns, non-significant.

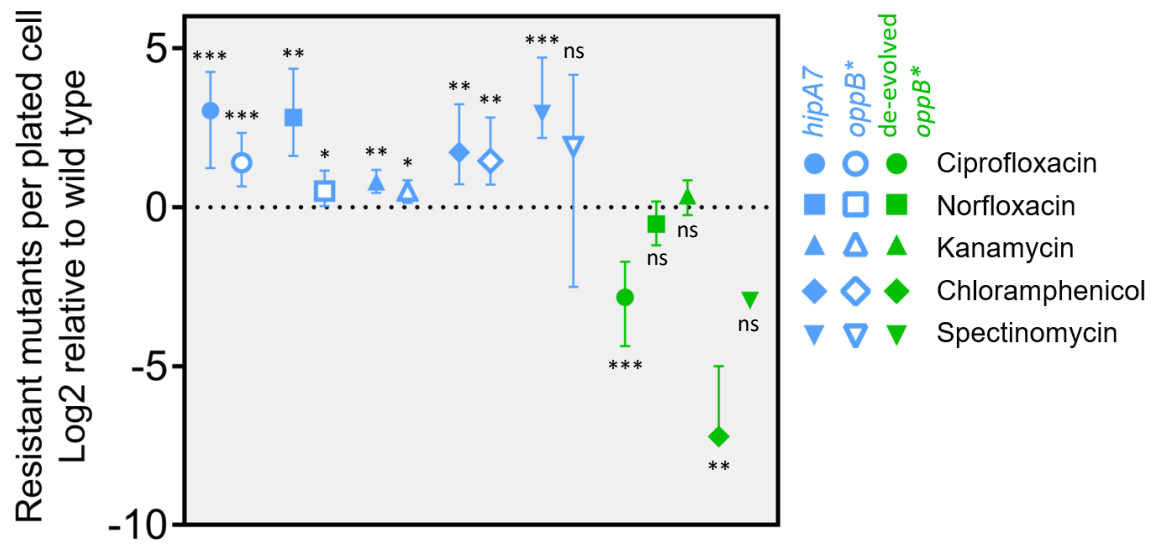


Figure S6. Emergence of antibiotic-resistant colonies on agar plates supplemented with various antibiotics. The number of resistant mutants per plated cell after 10 days on Mueller-Hinton agar supplemented with ciprofloxacin, norfloxacin, kanamycin, chloramphenicol and spectinomycin. Data points show the average number of resistant mutants per plated cell relative to the wild type. Error bars represent 95% c.i.'s (n=5-15), but are not shown when they are too large (de-evolved *oppB**; spectinomycin and chloramphenicol). For ciprofloxacin, these data are also represented in Fig. 2b of the main text. * $p \leq 0.05$, ** $p \leq 0.01$, *** $p \leq 0.001$, and ns, non-significant.

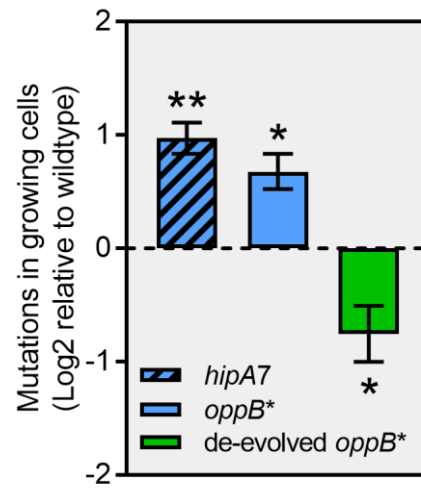


Figure S7. Mutation rate of growing cells. Mutations per cell per division, estimated by rifampicin-based fluctuation assays, relative to the wild type. Error bars show 95% c.i.'s (n=30-50), p-values are from one-sided *t*-tests. * $p \leq 0.1$, ** $p \leq 0.05$.

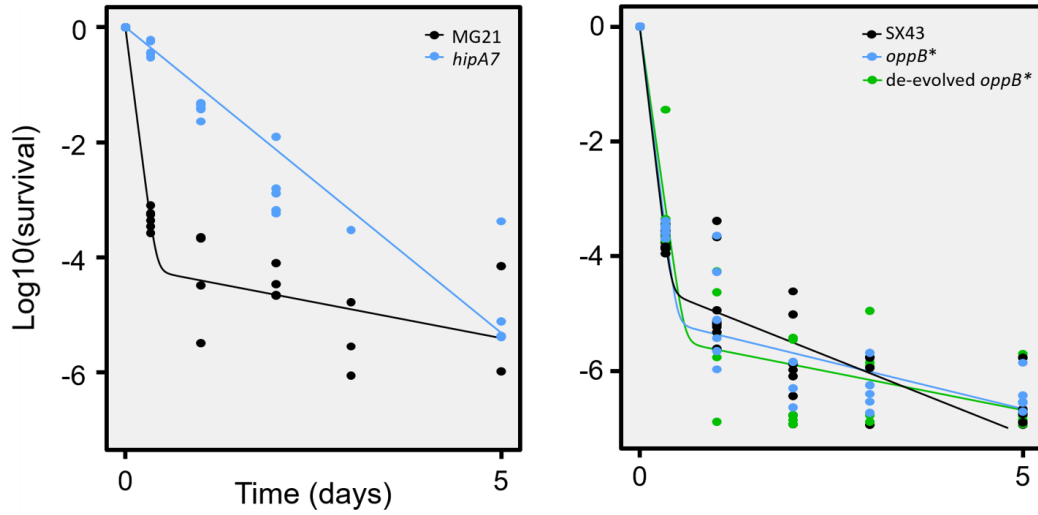


Figure S8. Time-kill kinetics of populations plated on ciprofloxacin-containing agar. Fraction of viable cells present on ciprofloxacin-containing agar plates. Individual data points are shown and lines represent average fits of biphasic exponential functions or a monophasic exponential function (*hipA7*). *oppB** does not show increased survival under these conditions. This is not unexpected, as its high-persistence phenotype was shown to be stationary phase specific and to disappear upon dilution in fresh medium (see ref 11 of the main text).

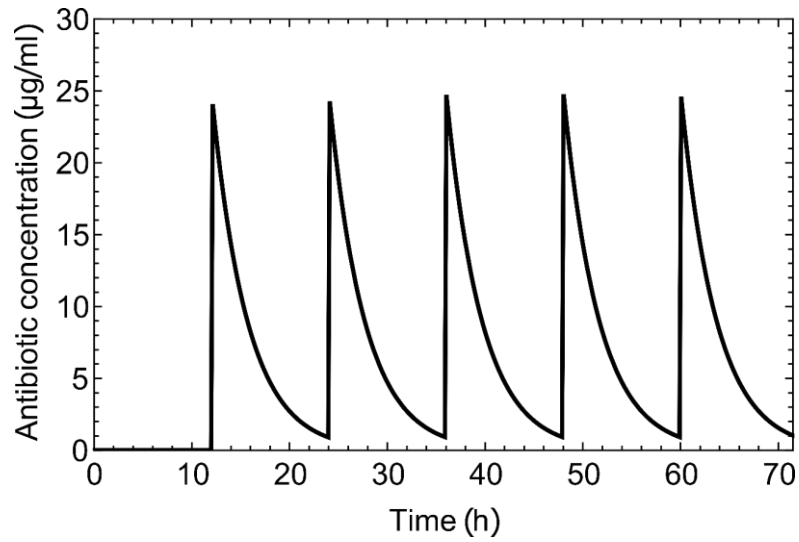


Figure S9. Antibiotic concentration profile. For the mathematical model of antibiotic treatment of a bacterial infection, we use a treatment schedule consisting of a peak dose of 25 µg/ml applied every 12 hours with the effective antibiotic concentration decaying exponentially with a half-life of 2.5 hours.

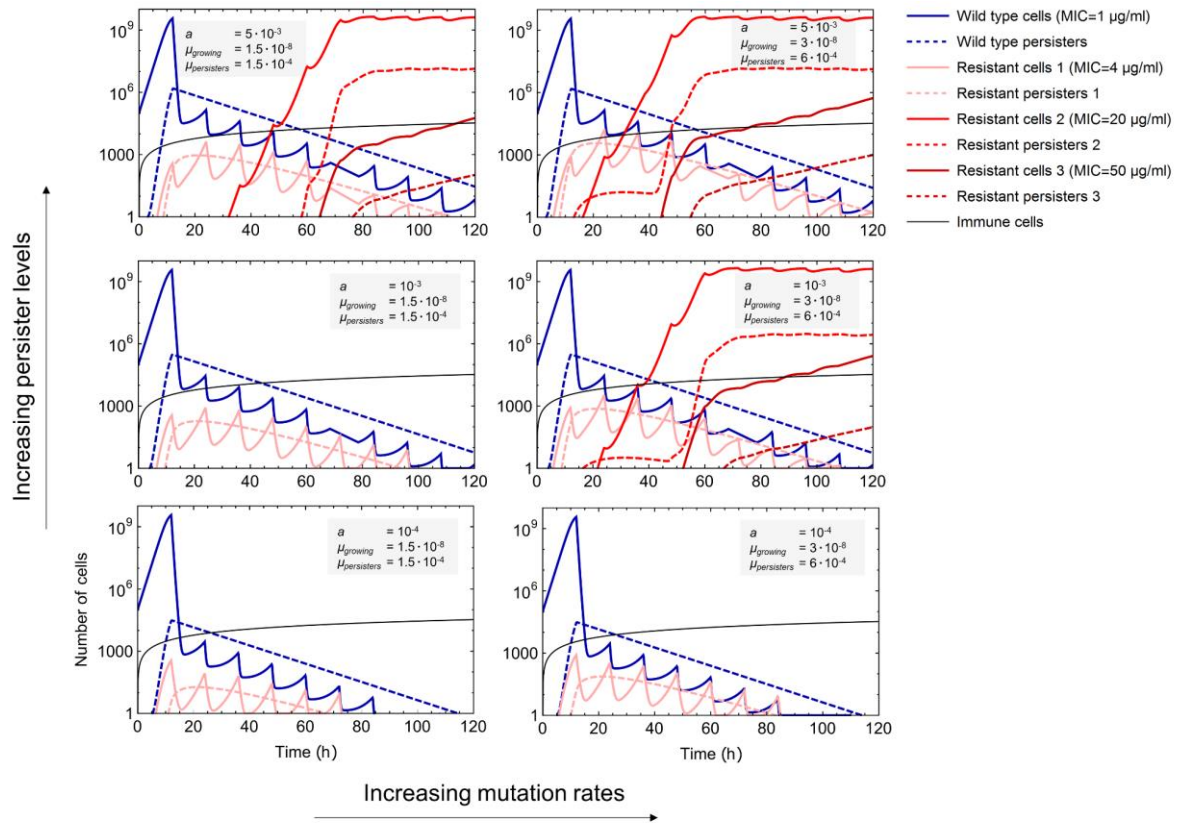


Figure S10. Population dynamics during antibiotic treatment of a bacterial infection. Separate dynamics of all bacterial populations and the population of immune effector cells is plotted for different sets of parameters. See Fig. 4 and the main text for details.

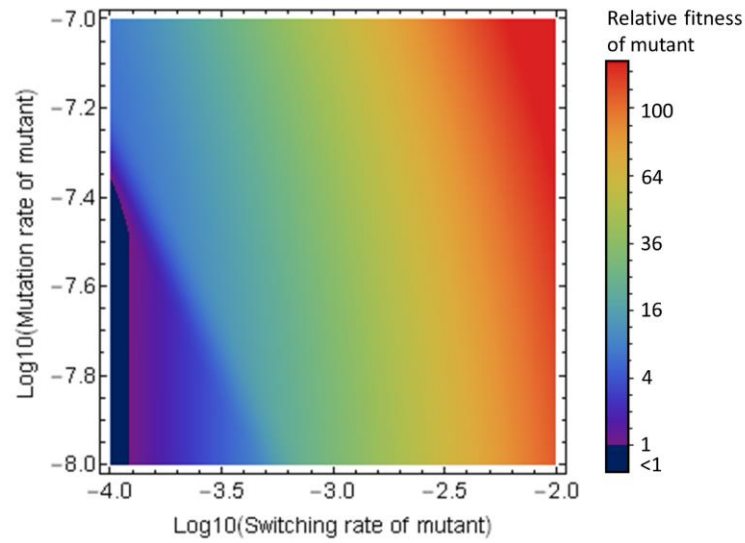


Figure S11. Selective advantage of rare mutants in a wild type population. Relative fitness of rare mutants with varying persister fractions (switching rates) and mutation rates in a wild type population, under conditions of an *in vivo* antibiotic treatment. The origin of the graph corresponds to the parameter values for the wild type. The relative fitness of a mutant is proportional to its switching rate as well as its mutation rate, and mutants with pleiotropically linked increased persistence and mutation rates show the highest selective advantage.

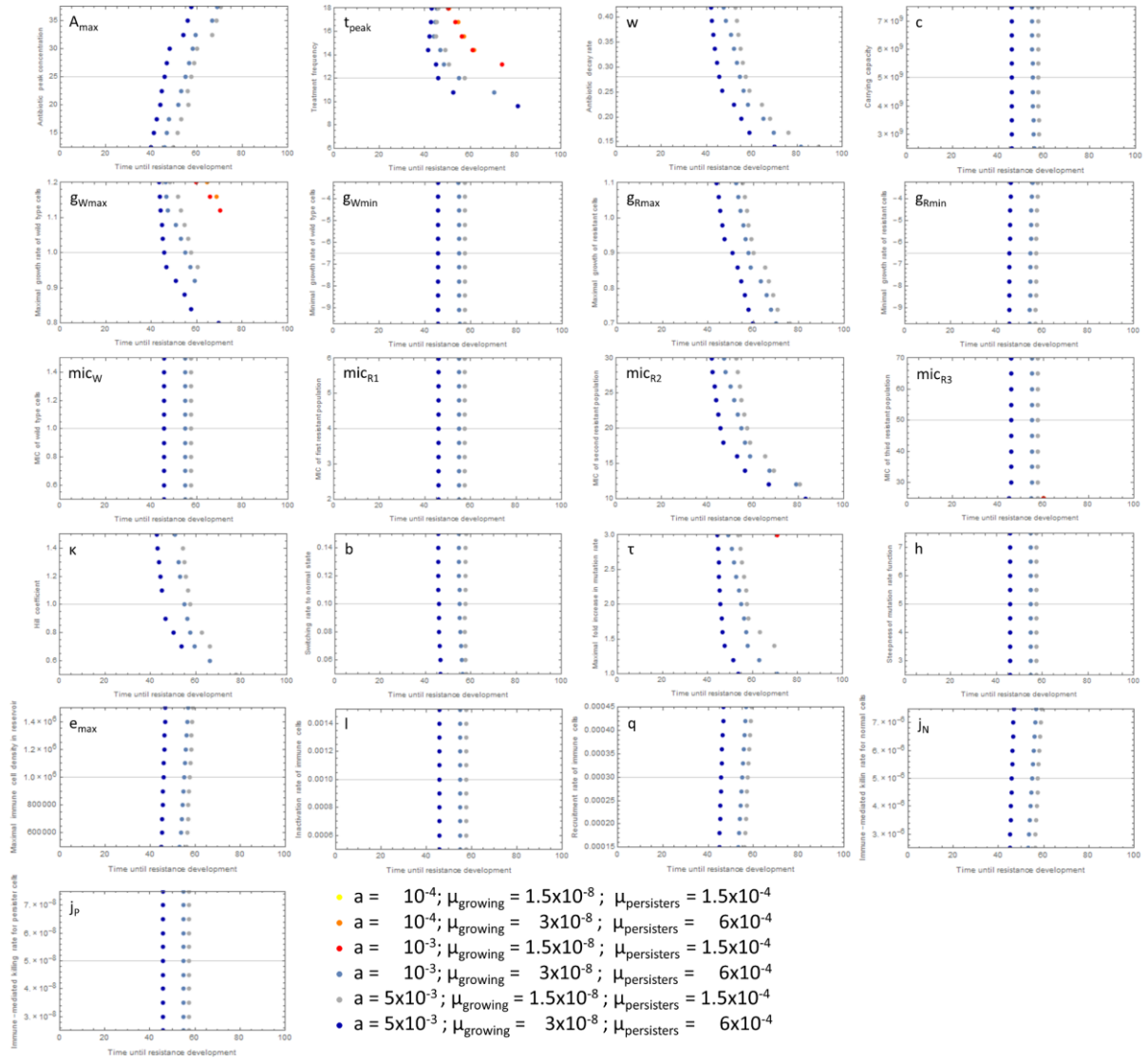


Figure S12. Sensitivity of the model predictions to parameter variations. Time until resistance development for strains with increasing persister fractions with or without associated increased mutation rates, plotted for different levels of each model parameter. When resistance did not develop within the course of the infection, no data point is plotted. Overall, our main model prediction, *i.e.* increased persistence and mutation rates jointly drive the evolution of resistance, was robust to variation in each individual model parameter.

Supplementary Tables

Table S1. Ciprofloxacin MICs ($\mu\text{g/ml}$) of the 20 ECOR isolates used in this study.

Strain	Ciprofloxacin MIC ($\mu\text{g/ml}$)
ECOR 1	0.008
ECOR 8	0.001
ECOR 15	0.008
ECOR 16	0.008
ECOR 21	0.008
ECOR 26	0.016
ECOR 30	0.016
ECOR 36	0.008
ECOR 37	0.008
ECOR 41	0.008
ECOR 42	0.004
ECOR 43	0.004
ECOR 47	0.004
ECOR 49	0.004
ECOR 51	0.008
ECOR 57	0.008
ECOR 58	0.008
ECOR 59	0.016
ECOR 65	0.008
ECOR 70	0.008

Table S2. Parameter estimates with 95% confidence intervals from a biphasic exponential fit to time-kill curves of ECOR strains (see Figure S1).

Strain	Log10(persister fraction)		Mortality rate of persister cells		Mortality rate of normal cells	
	Estimate	Confidence interval	Estimate	Confidence interval	Estimate	Confidence interval
ECOR 1	-0.40	[-0.82, 0.03]	0.38	[0.27, 0.56]	1.43	[0.56, 3.63]
ECOR 8	-4.61	[-4.88, -4.34]	0.03	[0.00, 0.59]	8.00	[7.55, 8.47]
ECOR 15	-0.96	[-1.37, -0.54]	0.42	[0.30, 0.58]	4.23	[1.52, 11.72]
ECOR 16	-1.70	[-2.01, -1.39]	0.26	[0.18, 0.39]	4.89	[3.46, 6.92]
ECOR 21	-1.12	[-1.43, -0.81]	0.06	[0.01, 0.38]	2.65	[1.75, 4.01]
ECOR 26	-1.81	[-2.17, -1.45]	0.02	[0.00, 5.66]	3.19	[2.59, 3.94]
ECOR 30	0.67	[0.36, 0.98]	0.54	[0.45, 0.66]	12.11	[0.00, Inf]
ECOR 36	-1.94	[-2.26, -1.63]	0.29	[0.18, 0.47]	4.53	[3.74, 5.48]
ECOR 37	-1.19	[-1.50, -0.89]	0.61	[0.51, 0.73]	2.96	[2.24, 3.92]
ECOR 41	-2.50	[-2.77, -2.24]	0.46	[0.38, 0.57]	6.17	[5.33, 7.13]
ECOR 42	-3.40	[-3.68, -3.11]	0.61	[0.52, 0.72]	9.27	[7.72, 11.15]
ECOR 43	-2.40	[-2.66, -2.13]	0.64	[0.55, 0.74]	6.83	[5.54, 8.41]
ECOR 47	-0.26	[-0.71, 0.19]	0.45	[0.26, 0.77]	2.04	[0.56, 7.49]
ECOR 49	-0.59	[-1.08, -0.11]	0.93	[0.71, 1.22]	5.51	[0.03, 948.46]
ECOR 51	-0.36	[-0.81, 0.10]	0.14	[0.05, 0.43]	2.38	[0.18, 32.20]
ECOR 57	-0.01	[-0.33, 0.31]	0.41	[0.31, 0.53]	1.34	[0.19, 9.23]
ECOR 58	-1.54	[-2.08, -0.99]	0.06	[0.00, 5.82]	2.66	[1.97, 3.58]
ECOR 59	-4.42	[-4.71, -4.13]	0.29	[0.19, 0.42]	9.46	[8.87, 10.08]
ECOR 65	-2.17	[-2.53, -1.82]	0.06	[0.01, 0.55]	4.88	[4.05, 5.89]
ECOR 70	-0.32	[-0.71, 0.08]	0.76	[0.63, 0.90]	4.60	[0.01, 4144.76]

Table S3. Parameter estimates and 95% confidence intervals from a biphasic exponential fit to time-kill curves of persistence mutants (see Figure 2a of the main text).

Strain	Log10(persister fraction)		Mortality rate of persister cells		Mortality rate of normal cells	
	Estimate	Confidence interval	Estimate	Confidence interval	Estimate	Confidence interval
SX43	-2.26	[-2.44, -2.08]	-0.35	[-0.44, -0.27]	0.71	[0.66, 0.76]
<i>oppB</i>*	-1.33	[-1.51, -1.16]	-9.27	[-Inf, Inf]	0.63	[0.49, 0.77]
de-evolved <i>oppB</i>*	-2.63	[-2.81, -2.44]	-0.31	[-0.39, -0.22]	1.06	[0.99, 1.14]
MG21	-1.24	[-1.51, -0.97]	-0.28	[-0.38, -0.18]	0.59	[0.39, 0.78]
<i>hipA7</i>	-0.39	[-0.88, 0.10]	-0.29	[-0.45, -0.14]	0.23	[-0.41, 0.88]

Table S4. MICs ($\mu\text{g/ml}$) of *oppB**, de-evolved *oppB**, *hipA7* and their isogenic wild types for ciprofloxacin (CIP), norfloxacin (NOR), kanamycin (KAN), chloramphenicol (CHL), and spectinomycin (SPT).

Strain	CIP ($\mu\text{g/ml}$)	NOR ($\mu\text{g/ml}$)	KAN ($\mu\text{g/ml}$)	CHL ($\mu\text{g/ml}$)	SPT ($\mu\text{g/ml}$)
SX43	0.008	0.064	1	4	16
<i>oppB</i>*	0.008	0.064	1	4	16
de-evolved <i>oppB</i>*	0.008	0.064	1	4	32
MG21	0.008	0.064	2	4	16
<i>hipA7</i>	0.008	0.064	2	4	16

Table S5. Absolute mutation rates (μ) with standard deviations (s.d.) estimated with fluctuation assays of populations of persistence mutants and wild types. Mutation rates are expressed as the number of mutations per cell per division.

Strain	Ciprofloxacin selection		Rifampicin selection	
	μ mutations/cell/division	s.d.	μ mutations/cell/division	s.d.
SX43	1.01E-08	2.28E-09	1.13E-08	2.89E-09
<i>oppB</i>*	5.04E-08	1.43E-08	1.89E-08	3.59E-09
de-evolved <i>oppB</i>*	4.27E-09	2.26E-09	6.37E-09	1.67E-09
MG21	2.89E-09	1.12E-09	1.62E-08	2.92E-09
<i>hipA7</i>	7.97E-09	3.02E-09	3.19E-08	6.08E-09

Table S6. Overview of variables and parameters in the mathematical model of an *in vivo* bacterial infection. All parameter values are chosen to be within the range for *E. coli* and relevant antibiotics used to treat *E. coli* infections. Further justification of the chosen parameters is given in Supplementary Methods.

Variable	Description		
N_W	Wild type normal cells, cells/ml		
P_W	Wild type persister cells, cells/ml		
N_{R1}	Normal cells of first resistant population, cells/ml		
P_{R1}	Persister cells of first resistant population, cells/ml		
N_{R2}	Normal cells of second resistant population, cells/ml		
P_{R2}	Persister cells of second resistant population, cells/ml		
N_{R3}	Normal cells of third resistant population, cells/ml		
P_{R3}	Persister cells of third resistant population, cells/ml		
E	Activated immune cells, cells/ml		
A	Antibiotic concentration, $\mu\text{g/ml}$		
Parameter	Description	Value	Reference
<i>Antibiotic treatment</i>			
$A_{\max}$	Antibiotic peak concentration, $\mu\text{g/ml}$	25	7
t_{peak}	Antibiotic dosing interval, h	12	7
w	Antibiotic decay rate, h^{-1}	0.27726	7
<i>Growth, killing, and persister formation</i>			
c	Carrying capacity, cells/ml	5×10^9	this study
$g_{W\max}$	Maximum growth rate of wild type cells, h^{-1} per cell	1	6
$g_{W\min}$	Minimum growth rate (death rate) of wild type cells, h^{-1} per cell	-6.5	6
$g_{R\max}$	Maximum growth rate of resistant cells, h^{-1} per cell	0.9	6
$g_{R\min}$	Minimum growth rate (death rate) of resistant cells, h^{-1} per cell	-6.5	6
miC_W	MIC of wild type cells, $\mu\text{g/ml}$	1	7
miC_{R1}	MIC of first resistant population, $\mu\text{g/ml}$	4	7
miC_{R2}	MIC of second resistant population, $\mu\text{g/ml}$	20	7
miC_{R3}	MIC of third resistant population, $\mu\text{g/ml}$	50	7
κ	Hill coefficient	1	6
$a_{\max}$	Maximum switching rate to persister state, h^{-1} per cell	$10^{-7} - 1$	1, 5
b	Switching rate to normal state, h^{-1} per cell	0.1	1, 5
<i>Mutation to resistance</i>			
m	Mutation rate to resistance in normal cells, mutations per division	$10^{-10} - 10^{-5}$	10, this study
τ	Maximum fold increase in mutation rate	2	10, this study
h	Curve steepness of mutation rate	5	9
μ_p	Mutation rate in persister cells, mutations h^{-1}	$10^{-6} - 10^{-1}$	11, this study
<i>Innate immune response</i>			
$e_{\max}$	Maximum immune cell density in the reservoir, number of cells	10^6	12
l	Inactivation rate of immune cells, h^{-1}	10^{-3}	12
q	Recruitment rate of immune cells, h^{-1}	3×10^{-4}	12
j_N	Immune-mediated killing rate for normal cells, h^{-1} per immune cell	5×10^{-6}	12
j_P	Immune-mediated killing rate for persister cells, h^{-1} per immune cell	5×10^{-8}	12

Supplementary References

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